EEEN20054 – Bending of beams

- This report should not exceed 6 pages; formatting settings should not be changed.

Name and ID number:	Itunuoluwa Oyewole 11377069

6. Experiment 1: Simply supported beam

6.2. Measuring reaction forces

[20 marks]

Table 6.2.1

	Left load cell		Right load cell		Calculated	
Apply 10 N to:	**Dial gauge (mm)**	**Reaction force (N)**	**Dial gauge (mm)**	**Reaction force (N)**	Left reaction force (N)	Right reaction force (N)
Hanger 1	0.44	4.4	0.45	4.5	5	5
Hanger 2	0.22	2.2	0.65	6.5	2.5	7.5
Hangers 1 and 2	0.66	6.6	1.11	11.1	7.5	12.5
Hanger 3	1.10	11	-0.24	-2.4	12.5	-2.5
Hangers 1 and 3	1.51	15.1	0.25	2.5	17.5	2.5

Comments:

The measured reaction forces from the load cell are approximately the calculated theoretical values. Small discrepancies could be due to friction in the load cell or dial gauges mechanism (despite tapping) and the metals weight. Or errors in the positioning of the hangers and not zeroing the load cells before the weights were added
The negative reading from hanger 3 is due to hanger 3 being outside the left support creasting a negative reaction force.

6.3. Measuring beam deflection

[50 marks]

Table 6.3.1

$I_y = 4.15x10^{-10}$	**Mild Steel**		**Brass**		**Aluminium**	
Applied load (N)	**Measured deflection (mm)**	Calculated deflection (mm)	**Measured deflection (mm)**	Calculated deflection (mm)	**Measured deflection (mm)**	Calculated deflection (mm)
5	0.83	0.84	2.03	1.67	2.29	2.04
10	1.75	1.67	4.05	3.34	4.58	4.08
15	2.62	2.51	6.09	5.01	6.86	6.12
20	3.49	3.34	8.05	6.68	9.10	8.16
25	4.37	4.18	10.09	8.35	11.32	10.20
30	5.26	5.01	12.09	10.02	13.62	12.24
35	6.13	5.85	14.16	11.69	15.84	14.27

Comments and plots (to include why measurements of deflection have been taken at this point along with the appropriate formula for calculating the deflection):

The formula used for calculating the deflection:

$$\boldsymbol{\omega}(\boldsymbol{x}) = \frac{\boldsymbol{F} * \boldsymbol{b} * \boldsymbol{x}}{\boldsymbol{6} * \boldsymbol{E} * \boldsymbol{I_y} * \boldsymbol{L}} (\boldsymbol{L^2} - \boldsymbol{b^2} - \boldsymbol{x^2})$$

Where w = defelction, F is the load, b = 19mm, h = 6.4mm, x = dial guage at 559mm, L = 1 m (length of metal), E = Young Modulus.

This was derived by integrating the standard formula of deflection:

$$E * I_y * \boldsymbol{\omega}'' = -\boldsymbol{M}(\boldsymbol{x}) = \frac{\boldsymbol{F} * \boldsymbol{b} * \boldsymbol{x}}{\boldsymbol{L}}$$

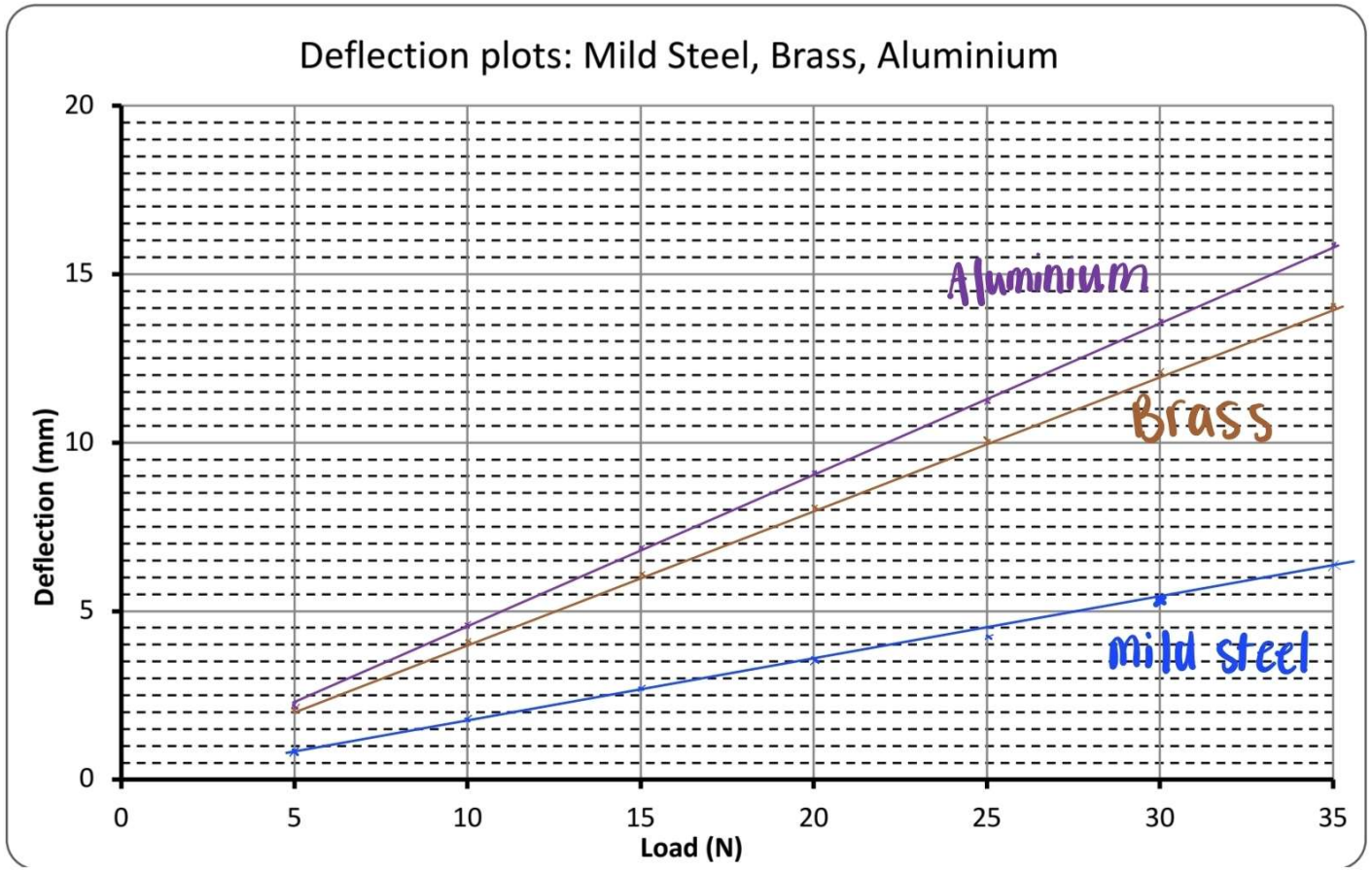
Where

$$I_y = \frac{b * h^3}{12} = \frac{0.019 * 0.0064^3}{12} = 4.15 \times 10^{-10}$$

obtaining the constants when x = 0 and x = a, (L = a + b).

Measurements where taken at x = 559mm because this is the point where maximum deflection occurs (ω'' = 0)

The results, as shown in the graph, show a linear relationship between the applied load and deflection for all three materials, confirming elastic behaviour within the tested ranges. The graph also clearly reflect the materials stiffness/Young Modulus. Mild steel is the stiffest with the smallest amount of defelection while aluminium is the least stiff showing the largest defelections. While there is linearity, there is a noticble discreprancy between the measured and calculated values, especially for Aluminium and Brass. The measured defelections are consistently higher than the calculated ones. This systematic and human error could be due to several reasons: slight movement in the supports, unalignment of beams and the incremental adding of weights during measurements introducing cumulative error rather than applying the full load directly. Lightly tapping helped reduce any frictional errors.


Deflection plots: Mild Steel, Brass, Aluminium
Deflection (mm)
Load (N)
0
5
10
15
20
25
30
35
Aluminum
Brass
mild steel

7. Experiment 2: Cantilever beam

7.2. Measuring beam deflection

[30 marks]

Table 7.2.1

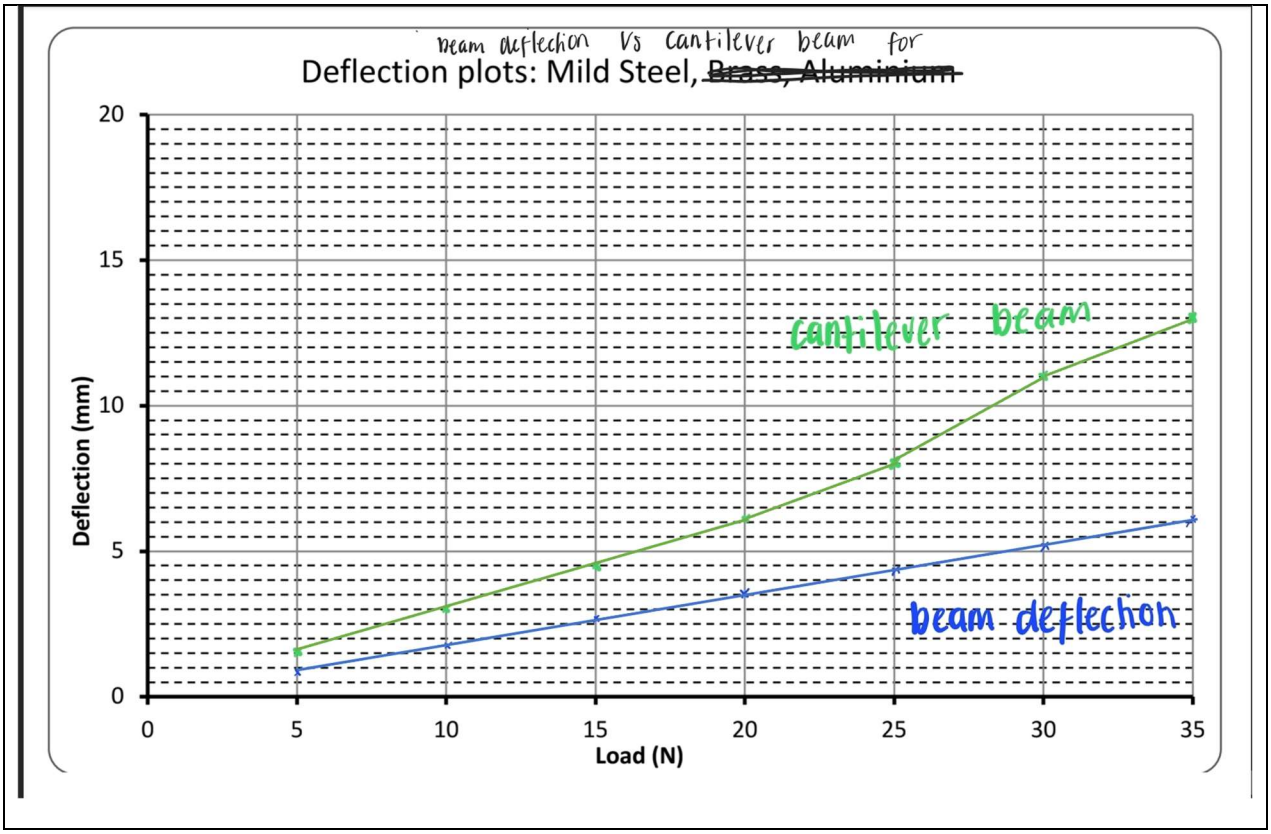
Second moment of area $I_y =$		4.15x10^-10
Load (N)	**Measured deflection (mm)**	Calculated deflection (mm)
5	1.43	1.23
10	2.94	2.45
15	4.46	3.67
20	6.23	4.90
25	8.11	6.12
30	11.15	7.34
35	12.95	8.57

Comments and plots (to include appropriate formula):

Equation used:

$$\boldsymbol{\omega}\,(\boldsymbol{x}) = \frac{\boldsymbol{F} * \boldsymbol{L}^{\boldsymbol{3}}}{\boldsymbol{3} * \boldsymbol{E} * \boldsymbol{I}_{\boldsymbol{y}}}$$

While the previous experiment exhibited linear behaviour, the cantilever beam results showed increasing non-linearity at higher loads. This error has several sources: the load hanger and the dial guage not being in the same position, the hanger not being fixed and the incremental adding of weights during measurements introducing cumulative error rather than applying the full load directly.


beam deflection Vs cantilever beam for
Deflection plots: Mild Steel, ~~Brass, Aluminium~~
Deflection (mm)
Load (N)
0
5
10
15
20
25
30
35
cantilever beam
beam deflection